

Supplementary for Solow Growth Model

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Abstract

In ch11, we assumed that labor is a constant, it wouldn't change from time to time. So we didn't consider the growth rate of labor (population). But now, we should consider that the labor would grow by a fixed rate n

Assumption

1. New classical production function,

$$Y = F(L, K); F_L > 0, F_{LL} < 0, F_K > 0, F_{KK} < 0$$

2. There is no money and government in the economy
3. Labor growth rate is fixed at rate n

Model

$$Y_t = F(L_t, K_t) \rightarrow \frac{Y_t}{L_t} = F\left(\frac{L_t}{L_t}, \frac{K_t}{L_t}\right) \rightarrow y_t = F\left(1, \frac{K_t}{L_t}\right) \rightarrow y_t = f(k_t)$$

1. $I_t = S_t$
2. $S_t = sY_t$
3. $\dot{K} = I_t - \delta K_t$

the third relation is the continuous case of $K_{t+1} = (1 - \delta)K_t + I_t$, where $\dot{K} = \frac{\partial K}{\partial t}$ is the continuous form of $K_{t+1} - K_t$.

Combine the above equations, we have

$$\dot{K}_t = sY_t - \delta K_t$$

divide both sides by N_t

$$\frac{\dot{K}_t}{N_t} = sy_t - \delta k_t$$
$$k_t = \frac{K_t}{N_t} \rightarrow \ln k_t = \ln K_t - \ln N_t$$

take partial derivative with respect to t on both sides

$$\frac{\dot{k}_t}{k_t} = \frac{\dot{K}_t}{K_t} - \frac{\dot{N}_t}{N_t} = \frac{\dot{K}_t}{K_t} - n; \quad \dot{k}_t = \frac{\partial k_t}{\partial t}$$

multiply both sides by k_t

$$\dot{k}_t = \frac{\dot{K}_t}{K_t} \frac{K_t}{N_t} - nk_t = \frac{\dot{K}_t}{N_t} - nk_t$$

then we get $\frac{\dot{K}_t}{N_t} = \dot{k}_t + nk_t$. Then, plug $\frac{\dot{K}_t}{N_t}$ into $\frac{\dot{K}_t}{N_t} = sy_t - \delta k_t$

$$\dot{k}_t + nk_t = sy_t - \delta k_t \rightarrow \dot{k}_t = sy_t - (n + \delta)k_t$$

We get the **fundamental accumulation equation** $\dot{k}_t = sy_t - (n + \delta)k_t$, when $\dot{k}_t = 0$, the economy reaches the steady state. Let's see a numerical example below.

Example 1 (We omitted the time index t for convenience) Consider the production function $Y = K^\alpha L^{1-\alpha}$, we can write the output per worker by dividing both sides by L

$$y = \frac{Y}{L} = K^\alpha L^{-\alpha} = \left(\frac{K}{L}\right)^\alpha = k^\alpha$$

Given the population growth rate, depreciation rate, and saving rate are n, δ , and s , respectively. According to the steady-state condition $\dot{k} = 0$ or $sy = (n + \delta)k$, we can derive

$$sy = (n + \delta)k \rightarrow sk^\alpha = (n + \delta)k \rightarrow \frac{s}{n + \delta} = k^{1-\alpha} \rightarrow k^* = \left(\frac{s}{n + \delta}\right)^{\frac{1}{1-\alpha}}$$

plug k^* into y , then we can get $y^* = \left(\frac{s}{n + \delta}\right)^{\frac{\alpha}{1-\alpha}}$. k^* is the steady state capital per worker, y^* is the steady-state output per worker. ■

Growth rates of Y_t and K_t when economy reach the steady-state are:

1.

$$k_t = \frac{K_t}{N_t} \rightarrow \frac{\dot{k}_t}{k_t} = \frac{\dot{K}_t}{K_t} - \frac{\dot{N}_t}{N_t} = \frac{\dot{K}_t}{K_t} - n \rightarrow \frac{\dot{K}_t}{K_t} = n$$

2.

$$y_t = \frac{Y_t}{N_t} \rightarrow \frac{\dot{y}_t}{y_t} = \frac{\dot{Y}_t}{Y_t} - \frac{\dot{N}_t}{N_t} = \frac{\dot{Y}_t}{Y_t} - n \rightarrow \frac{\dot{Y}_t}{Y_t} = n$$

So, the growth rates of Y_t and K_t are equal to the population growth rate, which is an exogenous variable. This is why the solow growth model is also called **exogenous growth model**.

Golden Rule

Definition 0.1. *The level of capital associated with the value of the saving rate that yields the highest level of consumption in steady state is known as the golden rule level of capital.*

Let's see a numerical example to understand the golden rule.

Example 2 Following from Example 1, since we want to maximize the consumption by choosing the saving rate (or capital per worker), we can write the representative consumer utility function as

$$U = (1 - s)f(k)$$

the utility function is measured by consumption, which is our main goal. Now, we can write down the optimal problem

$$\begin{aligned} \max_{k_t} \quad & U = (1 - s)f(k) \\ \text{s.t.} \quad & sf(k) = (n + \delta)k \end{aligned}$$

Due to the fact that $y = f(k) = k^\alpha$, we can rewrite the above optimal problem in the reduced form as,

$$\max \quad U = k^\alpha - (n + \delta)k$$

f.o.c.

$$\begin{aligned} \alpha k^{\alpha-1} &= n + \delta \rightarrow k^{\alpha-1} = \left(\frac{n + \delta}{\alpha}\right) \rightarrow k^* = \left(\frac{n + \delta}{\alpha}\right)^{\frac{1}{\alpha-1}} \\ s\left(\frac{n + \delta}{\alpha}\right)^{\frac{\alpha}{\alpha-1}} &= (n + \delta)\left(\frac{n + \delta}{\alpha}\right)^{\frac{1}{\alpha-1}} \rightarrow s\left(\frac{n + \delta}{\alpha}\right) = n + \delta \rightarrow s = \alpha \end{aligned}$$

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